

Rapid HIV-1 genotyping assay for the detection of capsid mutations.

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Human immunodeficiency virus (HIV) capsid is one of the most recent viral proteins successfully targeted for the development of antiretrovirals. Lenacapavir is a first in class HIV-1 capsid inhibitor that was recently approved for the treatment of highly treatment-experienced people with HIV in combination with other anti-HIV drugs. Owing to the novelty of the viral target, methods to characterize the potential resistance-associated mutations present in capsid upon treatment failure have not been fully established yet. Here, we describe a rapid and simple method to amplify capsid fragments and to determine their sequence from various clinical samples including diverse HIV-1 subtypes. These methods could easily be implemented in laboratories, including hospital laboratories often caring for this patient population.

Lenacapavir: A Novel, Potent, and Selective First-in-Class Inhibitor of HIV-1 Capsid Function Exhibits Optimal Pharmacokinetic Properties for a Long-Acting Injectable Antiretroviral Agent.

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Lenacapavir (LEN) is a picomolar first-in-class capsid inhibitor of human immunodeficiency virus type 1 (HIV-1) with a multistage mechanism of action and no known cross resistance to other existing antiretroviral (ARV) drug classes. LEN exhibits a low aqueous solubility and exceptionally low systemic clearance following intravenous (IV) administration in nonclinical species and humans. LEN formulated in an aqueous suspension or a PEG/water solution formulation showed sustained plasma exposure levels with no unintended rapid drug release following subcutaneous (SC) administration to rats and dogs. A high total fraction dose release was observed with both formulations. The long-acting pharmacokinetics (PK) were recapitulated in humans following SC administration of both formulations. The SC PK profiles displayed two-phase absorption kinetics in both animals and humans with an initial fast-release absorption phase, followed by a slow-release absorption phase. Noncompartmental and compartmental analyses informed the LEN systemic input rate from the SC depot and exit rate from the body. Modeling-enabled deconvolution of the input rates from two processes: absorption of the soluble fraction (minor) from a direct fast-release process leading to the early PK phase and absorption of the precipitated fraction (major) from an indirect slow-release process leading to the later PK phase. LEN SC PK showed flip-flop kinetics due to the input rate being substantially slower than the systemic exit rate. LEN input rates via the slow-release process in humans were slower than those in both rats and dogs. Overall, the combination of high potency, exceptional stability, and optimal release rate from the injection depot make LEN well suited for a parenteral long-acting formulation that can be administered once up to every 6 months in humans for the prevention and treatment of HIV-1.

Resistance Analyses in Highly Treatment-Experienced People With Human Immunodeficiency Virus (HIV) Treated With the Novel Capsid HIV Inhibitor Lenacapavir.

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Lenacapavir (LEN) is a first-in-class inhibitor of human immunodeficiency virus type 1 (HIV-1) capsid function in clinical development for the treatment of heavily treatment-experienced (HTE) people with

HIV (PWH) harboring multidrug resistance (MDR) in combination with an optimized background regimen (OBR). Here we describe resistance analyses conducted in the pivotal phase 2/3 CAPELLA study. CAPELLA enrolled viremic HTE PWH with resistance to ≥ 3 of 4 of the main antiretroviral (ARV) classes and resistance to ≥ 2 ARV drugs per class. Baseline resistance analyses used commercial assays (HIV-1 protease, reverse transcriptase, integrase genotypic/phenotypic tests). Postbaseline resistance was evaluated in participants experiencing virologic failure. At baseline, 46% of participants had resistance to the 4 main ARV drug classes, with one-third of participants having exhausted all drugs from ≥ 3 of the 4 main ARV classes. Treatment with LEN + OBR for 26 weeks led to viral suppression in 81% of participants. Postbaseline resistance mutations to lenacapavir occurred in 8 participants (6 with M66I, 1 with K70H, 1 with Q67H + K70R) who were receiving unintended functional LEN monotherapy at the time of resistance selection. LEN added to OBR led to high efficacy in this HTE patient population with MDR but could select for resistance when used unintentionally as functional monotherapy.

Structural and Mechanistic Bases for Resistance of the M66I Capsid Variant to Lenacapavir.

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Lenacapavir (LEN) is the first in class viral capsid protein (CA) targeting antiretroviral for treating multi-drug-resistant HIV-1 infection. Clinical trials and cell culture experiments have identified resistance associated mutations (RAMs) in the vicinity of the hydrophobic CA pocket targeted by LEN. The M66I substitution conferred by far the highest level of resistance to the inhibitor compared to other RAMs. Here we investigated structural and mechanistic bases for how the M66I change affects LEN binding to CA and viral replication. The high-resolution X-ray structure of the CA(M66I) hexamer revealed that the β -branched side chain of Ile66 induces steric hindrance specifically to LEN thereby markedly reducing the inhibitor binding affinity. By contrast, the M66I substitution did not affect binding of Phe-Gly (FG)-motif-containing cellular cofactors CPSF6, NUP153, or SEC24C, which engage the same hydrophobic pocket of CA. In cell culture the M66I variant did not acquire compensatory mutations or replicate in the presence of LEN. Analysis of viral replication intermediates revealed that HIV-1 (M66I CA) predominantly formed correctly matured viral cores, which were more stable than their wildtype counterparts. The mutant cores stably bound to the nuclear envelope but failed to penetrate inside the nucleus. Furthermore, the M66I substitution markedly altered HIV-1 integration targeting. Taken together, our findings elucidate mechanistic insights for how the M66I change confers remarkable resistance to LEN and affects HIV-1 replication. Moreover, our structural findings provide powerful means for future medicinal chemistry efforts to rationally develop second generation inhibitors with a higher barrier to resistance. Lenacapavir (LEN) is a highly potent and long-acting antiretroviral that works by a unique mechanism of targeting the viral capsid protein. The inhibitor is used in combination with other antiretrovirals to treat multi-drug-resistant HIV-1 infection in heavily treatment-experienced adults. Furthermore, LEN is in clinical trials for preexposure prophylaxis (PrEP) with interim results indicating 100 % efficacy to prevent HIV-1 infections. However, one notable shortcoming is a relatively low barrier of viral resistance to LEN. Clinical trials and cell culture experiments identified emergent resistance mutations near the inhibitor binding site on capsid. The M66I variant was the most prevalent capsid substitution identified in patients receiving LEN to treat multi-drug resistant HIV-1 infections. The studies described here elucidate the underlying mechanism by which the M66I substitution confers a marked resistance to the inhibitor. Furthermore, our structural findings will aid future efforts to develop the next generation of capsid inhibitors with enhanced barriers to resistance.

Long-acting lenacapavir acts as an effective preexposure prophylaxis in a rectal SHIV challenge macaque model.

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Long-acting antiretroviral agents for preexposure prophylaxis (PrEP) represent a promising new alternative to daily oral regimens for HIV prevention. Lenacapavir (LEN) is a first-in-class long-acting capsid inhibitor approved for the treatment of HIV-1 infection. Here, we assessed the efficacy of LEN for PrEP using a single high-dose simian-human immunodeficiency virus (SHIV) rectal challenge macaque model. In vitro, LEN showed potent antiviral activity against SHIV, as it did for HIV-1. In macaques, a single subcutaneous administration of LEN demonstrated dose proportional increases in and durability of drug plasma levels. A high-dose SHIV inoculum for the PrEP efficacy evaluation was identified via virus titration in untreated macaques. LEN-treated macaques were challenged with high-dose SHIV 7 weeks after drug administration, and the majority remained protected from infection, as confirmed by plasma PCR, cell-associated proviral DNA, and serology testing. Complete protection and superiority to the untreated group was observed among animals whose LEN plasma exposure exceeded its model-adjusted clinical efficacy target at the time of challenge. All infected animals had subprotective LEN concentrations and showed no emergent resistance. These data demonstrate effective SHIV prophylaxis in a stringent macaque model at clinically relevant LEN exposures and support the clinical evaluation of LEN for HIV PrEP in humans.

US cost-utility model of lenacapavir plus optimized background regimen (OBR) vs fostemsavir plus OBR and ibalizumab plus OBR for people with HIV with multidrug resistance.

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Heavily treatment-experienced (HTE) people with HIV (PWH) have limited treatment options owing to multidrug resistance (MDR). Lenacapavir (LEN) is indicated, in combination with other antiretrovirals, for the treatment of adults with MDR HIV-1 experiencing failure of their current antiretroviral regimen because of resistance, intolerance, or safety considerations. To evaluate the cost-utility of LEN in combination with an optimized background regimen (OBR) vs alternative recently approved treatments for HTE PWH, fostemsavir (FTR)+OBR and ibalizumab (IBA)+OBR, for the treatment of PWH with MDR, from a mixed US health care payer perspective. A Markov state-transition model with a lifetime time horizon was developed. Transition probabilities between viral load categories were based on individual participant data from the CAPELLA trial for LEN+OBR and on relative efficacy parameters obtained from indirect treatment comparisons for comparators. Health state utilities were sourced from the literature. Costs included drug acquisition costs, drug administration costs, disease management costs, adverse event costs, AIDS-related event costs, and treatment switching costs and were sourced from red book costs, Medicare and Medicaid fees, and the literature. Costs and outcomes were discounted at 3% annually. The model was used to estimate total and incremental costs, life-years (LYs), quality-adjusted life-years (QALYs), and incremental cost-effectiveness ratios. A deterministic and a probabilistic sensitivity analysis, as well as scenario analyses, were performed to address elements of uncertainty in the model and to explore the robustness of the results. Over a lifetime time horizon, LEN+OBR was associated with the highest absolute QALYs (9.41) and the greatest number of LYs (12.09) compared with FTR+OBR (QALYs: 8.75; LYs: 11.26) and IBA+OBR (QALYs: 8.36; LYs: 10.78). LEN+OBR was also

associated with the lowest total lifetime costs of the 3 interventions (LEN+OBR: \$1,441,122 [US dollars]; FTR+OBR: \$1,504,986; IBA+OBR: \$1,524,396) and therefore was dominant over both comparators in the base case. LEN+OBR remained dominant vs FTR+OBR and IBA+OBR across the range of scenarios tested and LEN+OBR had a 99% probability of being cost-effective compared with FTR+OBR and IBA+OBR in the probabilistic sensitivity analysis at a willingness-to-pay threshold of \$50,000/QALY. This economic evaluation demonstrated that LEN+OBR provides meaningful increases in QALYs and LYs, and is dominant over a lifetime time horizon, compared with FTR+OBR and IBA+OBR for the treatment of PWH with MDR in the United States.

Lenacapavir: Patient and healthcare provider perceptions and the potential role for a twice-yearly injectable HIV treatment.

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The twice-yearly, long-acting lenacapavir (LA LEN) antiretroviral therapy (ART), when combined with an optimised background regimen, provides a subcutaneous injectable treatment option for people with HIV. This study aimed to understand the preferences, barriers and facilitators for uptake and implementation of LA LEN, with a view to informing clinical implementation. In-depth qualitative interviews and focus groups with purposively sampled people with HIV and healthcare workers (HCWs) from UK HIV services were conducted. Transcripts were analysed using summative and conventional content analysis. Thirty-four people with HIV with varied ART experience were recruited from two HIV services. Participants included 22 (65%) identifying as cisgender men and 12 (35%) identifying as cisgender women; median age was 55 years (range 26-76 years). Fourteen HIV HCWs took part in three focus groups. Four key themes and 12 subthemes were identified: LA LEN as a treatment option; LA LEN versus oral ART; switching considerations; and administration of LA LEN. The majority (88%) of people with HIV were interested in switching to LA LEN if offered. Preference was markedly reduced if an oral ART pairing was required. Convenience of the dosing schedule, reduced pill burden and issues around stigma were reasons for interest in LA LEN, but concerns regarding efficacy, dosing interval windows, monitoring and side effects were described. HCWs felt the benefit of LA LEN was as a treatment option for those with adherence issues, drug resistance and a high pill burden. Broader use of LA LEN raised concerns over drug resistance, delivery capacity and storage. LA LEN was viewed as a preferable treatment choice for many people with HIV, provided an all-injectable regimen was available. Feasibility assessments for provision of injectable ART and research on its potential for self-administration are needed.

Pharmacokinetics, Disposition, and Biotransformation of [14C]Lenacapavir, a Novel, First-in-Class, Selective Inhibitor of HIV-1 Capsid Function, in Healthy Participants Following a Single Intravenous Infusion.

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Lenacapavir (LEN) is a novel, first-in-class, multistage, selective inhibitor of human immunodeficiency virus type 1 (HIV-1) capsid function recently approved for the treatment of HIV-1 infection in heavily treatment-experienced adults with multidrug-resistant HIV-1 infection. The purpose of this multicohort study was to evaluate the pharmacokinetics, metabolism, excretion, safety, and tolerability of LEN following a single intravenous (IV) infusion of 10 mg LEN or 20 mg [14C]LEN in healthy participants.

Twenty-one healthy adult participants were enrolled into the study and received either a single IV dose of 10 mg LEN (n = 8 active, n = 3 placebo; cohort 1) or a single IV dose of 20 mg [14C]LEN containing 200 μ Ci (n = 10; cohort 2). Blood, urine, and feces samples (when applicable) were collected after dosing, and radioactivity (cohort 2) was assessed using liquid scintillation counting in both plasma and excreta. LEN in plasma was quantified by liquid chromatography (LC) tandem mass spectroscopy (MS/MS) method bioanalysis. Metabolite profiling in plasma and excreta were performed using LC-fraction collect (FC)-high-resolution MS and LC-FC-accelerator mass spectrometry in plasma. Between the 10 mg and 20 mg doses of LEN, the observed plasma exposure of LEN doubled, while the elimination half-life was similar. Following administration of 20 mg [14C]LEN (200 μ Ci), the mean cumulative recovery of [14C] radioactivity was 75.9% and 0.24% from feces and urine, respectively. The mean whole [14C] blood-to-plasma concentration ratio was 0.5-0.7, which showed a low distribution of LEN to red blood cells. Intact LEN was the predominant circulating species in plasma (representing 68.8% of circulating radioactivity), and no single metabolite contributed to > 10% of total radioactivity exposure through 1176 h postdose. Similarly, intact LEN was the most abundant component (32.9% of administered dose; 75.9% of recovered dose) measured in feces, with metabolites accounting for trace amounts. These results suggest metabolism of LEN is not a primary pathway of elimination. Of the metabolites observed in the feces, the three most abundant metabolites were direct phase 2 conjugates (glucuronide, hexose, and pentose conjugates), with additional metabolites formed to a lesser extent via other pathways. The administered LEN IV doses were generally safe and well-tolerated across participants in this study. The results of this mass balance study indicated that LEN was majorly eliminated as intact LEN via the feces. The renal pathway played a minor role in LEN elimination (0.24%). In addition, no major circulating metabolites in plasma or feces were found, indicating minimal metabolism of LEN.

Can a unilateral total temporomandibular joint prosthesis affect the healthy contralateral temporomandibular joint? A systematic review.

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Temporomandibular joint (TMJ) prostheses are considered an important alternative for the treatment of severe end-stage TMJ disorders. However, unilateral TMJ replacement may affect the healthy contralateral TMJ. The purpose of this study was to systematically review the literature on the effects of unilateral total TMJ prosthesis placement on the contralateral healthy TMJ. The PubMed, Scopus, Web of Science, and Cochrane Library databases were searched for English-language articles published up to December 2023. Inclusion criteria encompassed clinical studies (randomized, prospective, retrospective, observational) that evaluated clinical and patient-reported outcomes after total unilateral TMJ prosthesis placement (both glenoid and mandibular components). The initial search identified 141 non-duplicate studies, of which eight remained after title and abstract reading. Four studies included only unilateral prostheses; the other four included bilateral interventions and/or control groups without any TMJ intervention. The studies reported on stock and custom prostheses, with custom prostheses being the most used. As the studies had different specific objectives, no pattern of data reporting was found and the research question could not be answered. Randomized clinical trials with standardized variables are required to achieve reliable conclusions. Furthermore, long-term follow-up is necessary to determine whether the function of the healthy TMJ is compromised.

Treatment of late-onset temporomandibular joint prosthesis infection by prosthesis

revision: a case report.

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Late-onset infection of an inserted temporomandibular joint prosthesis is a difficult complication to treat. Most treatment protocols for late prosthetic infections include device replacement. A 40-year-old female patient with an infected and exposed temporomandibular joint prosthesis presented 3 years after implant placement. The patient was treated with prosthesis revision including fistula coverage with a temporalis muscle flap and prolonged antibiotic therapy for 10 weeks. Since completion of treatment, the patient has been infection-free.
